Tianyuan (Chris) Liu

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Robotics Engineer | Robot Learning | Manipulation | Real-Robot Deployment

Professional Summary

Robotics PhD candidate with hands-on experience in **real-robot integration, learning-based control, and sim-to-real deployment**. Built perception and localization pipelines for **Unitree G1**, deployed reinforcement-learning-based manipulation on **xArm7**, and developed simulation and data workflows across humanoid and manipulator platforms. Combines robotics research with practical system debugging and deployment.

Technical Skills

Software / frameworks: Python, C++, ROS / ROS 2, PyTorch, Isaac Lab / Isaac Sim, Docker, HPC.

Robotics / ML: Robot perception, calibration, 6-DoF tracking, motion planning, residual PPO, imitation learning, teacher–student learning, sim-to-real.

Hardware / systems: Unitree G1, xArm7, UR5 / Robotiq, Walker S2, RB-Y1; FoundationPose, MoveIt, cuMotion, DDS.

Professional Experience

Samsung Research China — Robotics / Embodied AI Research Intern Aug 2026 – Present

- Contributed to **end-to-end real-robot deployment** of a **Unitree G1** humanoid box-handling pipeline spanning grasping, carrying, placement, and scanning.
- Built shared-frame **localization and calibration** workflows with trackers, plus real-time **6-DoF object tracking** using FoundationPose to support robot perception and control integration.
- Independently trained a **teacher–student whole-body control policy**; developed HiPHI / HoloSoma motion-retargeting workflows for humanoid data collection.
- Brought up a **Rainbow Robotics RB-Y1**, including base mobility and hand–eye calibration, within two days.

Zoellent Technology — Robotics AI Research Intern Jun 2026 – Aug 2026

- Built **Isaac Sim / Isaac Lab and ROS 2** manipulation workflows, including DAgger imitation learning and perception–planning–control integration for simulation and physical-robot tests.
- Investigated VLA, world-model, and multimodal methods for humanoid manipulation on UBTECH Walker S2.

Monash University — Research Assistant Nov 2022 – Mar 2023

- Co-developed a **vision–force–tactile closed-loop** control system for in-hand pivoting on a physical **UR5 / Robotiq**, combining force/torque feedback and tactile slip regulation.
- Demonstrated 90° cube rotation without lifting or translational slip; work contributed to an **IROS 2023** publication (second author).

Selected Robotics Projects

Contact-Rich Book Insertion — Residual Reinforcement Learning

Deakin PhD research | ICRA 2027 under review, first author

- Developed a phase-aware hybrid controller combining nominal task-space motion with **residual PPO** for contact correction and release timing; trained using **256 parallel Isaac Lab environments**.
- Deployed on a **physical xArm7** with ROS 2 and 20-Hz velocity control; insertion success increased from **26.7% to 63.3%** in 60 paired trials. In simulation, achieved 93.3% vs. 27.1% for nominal control at 3-mm clearance.

Execution-Aware Robotic Placement — Learning-Based Candidate Ranking

Deakin PhD research | ICRA 2027 under review, first author

- Built a **PointNet++-based** grasp-conditioned ranking system with joint planning/execution prediction heads, trained on 153k candidates from 30 objects and 1,235 scenes.
- Improved top-1 hit rate to **85.6% vs. 72.8%** baseline; transferred the frozen model to **xArm7 without retraining**, with 13 successful placements out of 16 executed cases.

Grasp-Dependent Placement Feasibility Prediction

Deakin PhD research | ACRA 2025, first author

- Built a grasp-conditioned feasibility model using inexpensive IK and collision labels; improved fixed-budget placement success to **87% vs. 74%** in RRT-Connect / PhysX validation.

Education

Deakin University — PhD in Robotics / Artificial Intelligence Nov 2023 – May 2027 (expected)

Thesis submission expected February 2027

The University of Melbourne — Master of Information Technology (AI) Jul 2021 – Jul 2023

Dean's List 2022 (top 5%).

The University of Melbourne — BSc (Mechatronics Engineering) Feb 2018 – Jul 2021

Selected Publications

- **T. Liu et al.**, "Hybrid Residual Reinforcement Learning for Contact-Rich Robotic Book Insertion," ICRA 2027 (under review). [arXiv]
- **T. Liu et al.**, "Execution-Aware Pre-Execution Ranking for Grasp-Conditioned Robotic Placement," ICRA 2027 (under review). [arXiv]
- **T. Liu et al.**, "Optimizing Robotic Placement via Grasp-Dependent Feasibility Prediction," ACRA 2025.
- **S. Xu, T. Liu et al.**, "Rotating Objects via in-Hand Pivoting Using Vision, Force and Touch," IROS 2023.